Stochastic Approximation Algorithms Applied to Optimal Execution

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Outline

1. Optimal split of orders across liquidity pools
   - Modelling and mean execution cost
   - Design of the stochastic algorithm
   - Numerical experiments
2. Optimal posting price of limit orders: learning by trading
   - Intensity of execution
   - Mean cost function and stochastic algorithm
   - Cost function and its derivative as expectations
   - Numerical experiments

Outline

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Modelling see [6]

The principle of a *Dark pool* is the following:

- It proposes a bid price with no guarantee of executed quantity at the occasion of an OTC transaction.
- Usually this price is lower than the bid price offered on the regular market.

So one can model the impact of the existence of N dark pools ($N \geq 2$) on a given transaction as follows:

- Let $V > 0$ be the random volume to be executed,
- Let $\theta_i \in (0, 1)$ be the *discount factor* proposed by the dark pool i.
- Let r_i denote the percentage of V sent to the dark pool i for execution.
- Let $D_i \geq 0$ be the quantity of securities that can be delivered (or mase available) by the dark pool i at price $\theta_i S$.

Cost of the executed order

The reminder of the order is to be executed on the regular market, at price S.

Then the **cost** C of the whole executed order is given by

$$\begin{aligned} C &= S\sum_{i=1}^{N}\theta_i \min\left(r_i V, D_i\right) + S\left(V - \sum_{i=1}^{N}\min\left(r_i V, D_i\right)\right) \\ &= S\left(V - \sum_{i=1}^{N}\rho_i \min\left(r_i V, D_i\right)\right) \end{aligned}$$

where

$$\rho_i = 1 - \theta_i \in (0,1), i = 1, \ldots, N.$$

Mean Execution Cost

Minimizing the mean execution cost, *given the price* S, amounts to solving the following **maximization problem**

$$\max \left\{ \sum_{i=1}^{N} \rho_i \mathbb{E}\left(S \min\left(r_i V, D_i\right)\right), r \in \mathcal{P}_N \right\} \tag{1}$$

where $\mathcal{P}_N := \left\{ r = (r_i)_{1 \le i \le N} \in \mathbb{R}_+^N \mid \sum_{i=1}^{N} r_i = 1 \right\}$.

It is then convenient to *include the price* S *into both random variables* V *and* D_i by considering

$$\widetilde{V} := V\,S \quad \text{and} \quad \widetilde{D}_i := D_i S$$

instead of V and D_i.

Outline

Optimal allocation of orders among N dark pools

We set for every $r = (r_1, \ldots, r_N) \in \mathcal{P}_N$,

$$\Phi(r_1, \ldots, r_N) := \sum_{i=1}^{N} \varphi_i(r_i).$$

where for every $i \in I_N = \{1, \ldots, N\}$,

$$\varphi_i(u) := \rho_i \mathbb{E}\left[\min\left(uV, D_i\right)\right], \quad u \in [0,1].$$

We can formally extend Φ on the whole affine hyperplan spanned by $\mathcal{P}_N$ *i.e.*

$$\mathcal{H}_N := \left\{ r = (r_1, \ldots, r_N) \in \mathbb{R}^N \mid \sum_{i=1}^{N} r_i = 1 \right\}.$$

The Lagrangian Approach

We aim at solving the following maximization problem

$$\max_{r\in\mathcal{P}_N} \Phi(r). \tag{2}$$

The Lagrangian associated to the sole affine constraint is

$$L(r,\lambda) = \Phi(r) - \lambda\left(\sum_{i=1}^{N} r_i - 1\right) \tag{3}$$

Thus

$$\forall i \in I_N, \quad \frac{\partial L}{\partial r_i} = \varphi_i'(r_i) - \lambda.$$

This suggests that any $r^* \in \arg\max_{\mathcal{P}_N} \Phi$ iff $\varphi_i'(r_i^*)$ is constant when i runs over I_N or equivalently if

$$\forall i \in I_N, \quad \varphi_i'(r_i^*) = \frac{1}{N}\sum_{j=1}^{N} \varphi_j'(r_j^*). \tag{4}$$

Design of the stochastic algorithm

Then using that

$$\forall i \in I_N, \quad \varphi_i'(r) = \rho_i \mathbb{E}\left[\mathbb{1}_{\{r_i V < D_i\}} V\right],$$

this yields that, if $\arg\max_{\mathcal{H}_N} \Phi = \arg\max_{\mathcal{P}_N} \Phi \subset int(\mathcal{P}_N)$, then

$$r^* \in \arg\max_{\mathcal{P}_N} \Phi$$
$$\Updownarrow$$
$$\forall i \in I_N, \quad \mathbb{E}\left[V\left(\rho_i \mathbb{1}_{\{r_i^* V < D_i\}} - \frac{1}{N}\sum_{j=1}^N \rho_j \mathbb{1}_{\{r_j^* V < D_j\}}\right)\right] = 0.$$

Consequently, this leads to the following **recursive zero search procedure**

$$r_i^{n+1} = r_i^n + \gamma_{n+1} H_i(r^n, Y^{n+1}),\ r^0 \in \mathcal{P}_N,\ i \in I_N, \tag{5}$$

Design of the stochastic algorithm

where for $i \in I_N$, every $r \in \mathcal{P}_N$, every $V > 0$ and every $D_1, \ldots, D_N \geq 0$,

$$H_i(r, Y) = V \left(\rho_i \mathbb{1}_{\{r_i V < D_i\}} - \frac{1}{N} \sum_{j=1}^{N} \rho_j \mathbb{1}_{\{r_j V < D_j\}} \right)$$

where $(Y^n)_{n \geq 1}$ is a sequence of random vectors with non negative components such that, for every $n \geq 1$ and $i \in I_N$, $(V^n, D_i^n) \stackrel{d}{=} (V, D_i)$.

Comment: The underlying idea of the algorithm is to reward the dark pools which outperform the mean of the N dark pools by increasing the allocated volume sent at the next step (and conversely).

For results on *a.s.* convergence, see [2, 3, 4] for i.i.d. setting and [5] for averaging setting.

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Numerical Tests

Two natural situations of interest can be considered *a priori*:

- *abundance*: $\mathbb{E}\,V \leq \sum_{i=1}^{N} \mathbb{E} D_i$,
- *shortage*: $\mathbb{E}\,V > \sum_{i=1}^{N} \mathbb{E} D_i$.

We define a reference strategy named "**oracle**" devised by an insider who knows all the values V^n and D_i^n before making his/her optimal execution requests to the dark pools. The "oracle" strategy yields a cost reduction of the execution denoted by CR^{oracle}.

We introduce indexes to measure the **performances** of our recursive allocation procedure as follows: the ratios between the relative cost reductions of our allocation algorithm and that of the oracle, *i.e.*

$$\frac{CR^{opti}}{CR^{oracle}}.$$

The pseudo-real data setting

V is the traded volume of a very liquid security – namely the asset BNP – during an 11 day period.

Then we selected the N most correlated assets (in terms of traded volumes) with the original asset, denoted S_i, $i = 1, \dots, N$.

The available volumes of each dark pool i have been modeled as follows

$$\forall 1 \le i \le N, \quad D_i := \beta_i \left((1 - \alpha_i) V + \alpha_i S_i \frac{\mathbb{E}\, V}{\mathbb{E} S_i} \right)$$

where

- $\alpha_i \in (0, 1)$, $i = 1, \dots, N$ are the recombining coefficients,
- β_i, $i = 1, \dots, N$ are some scaling factors,
- $\mathbb{E}\, V$ and $\mathbb{E} S_i$ stand for the empirical mean of the data sets of V and S_i.

Parameters

The shortage situation corresponds to $\sum_{i=1}^{N} \beta_i < 1$ since it implies

$$\mathbb{E}\left[\sum_{i=1}^{N} D_i\right] < \mathbb{E}\, V.$$

We will use the following parameters

$$N = 4, \quad \beta = (0.1, 0.2, 0.3, 0.2)' \quad \text{and} \quad \alpha = (0.4, 0.6, 0.8, 0.2)'$$

and the dark pool trading (rebate) parameters are set to

$$\rho = (0.01, 0.02, 0.04, 0.06)'.$$

Long-term optimization

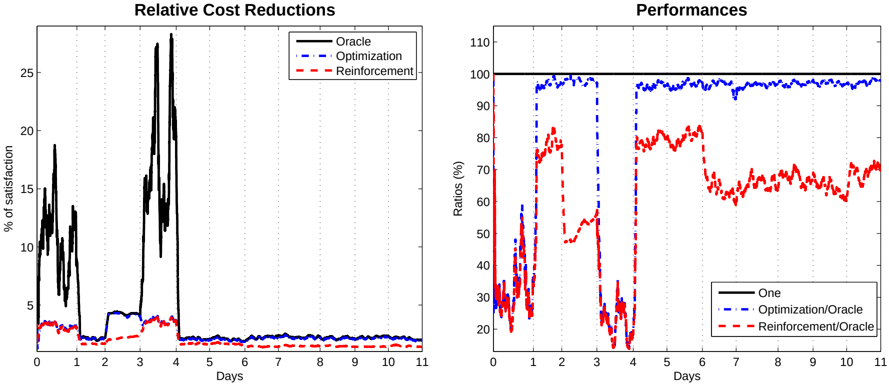

Figure: *Long term optimization*: Case $N = 4$, $\sum_{i=1}^{N} \beta_i < 1$, $0.2 \leq \alpha_i \leq 0.8$ and $r_i^0 = 1/N$, $1 \leq i \leq N$.

Daily resetting of the procedure

We reset the step γ_n at the beginning of each day and the satisfaction parameters and we keep the allocation coefficients of the preceding day.

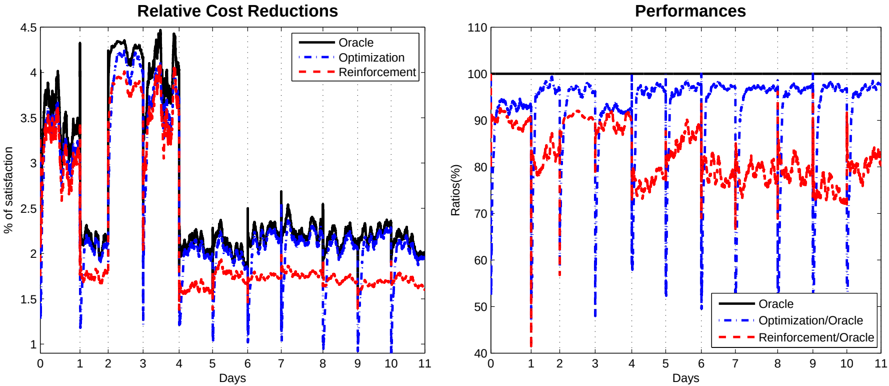

Figure: *Daily resetting of the algorithms parameters*: Case $N = 4$, $\sum_{i=1}^{N} \beta_i < 1$, $0.2 \leq \alpha_i \leq 0.8$ and $r_i^0 = 1/N$ $1 \leq i \leq N$.

Implementation

- Use the datasets on Paris Bourse: you have 4 assets (Bouygues, LVMH, Sanofi and Total). Select two weeks of data.
- Choose one of the asset for the volume you want to trade and the other to build the volumes on the dark pools.
- Fix an horizon T (in seconds) and compute the executed quantity on each period of T seconds executed for each assets (namely V and S_i).
- Choose the recombining coefficients α_i and the scaling factors β_i to be in the shortage setting and compute the volumes D_i.
- Choose ρ and implement both the stochastic algorithm and the oracle procedure. Print and comment your results.

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Modeling the process of execution, see [7]

We consider on a short period T a **Poisson process of “execution”** of buy orders

$$\left(N_t^{(\delta)}\right)_{0\leq t\leq T} \quad \text{with intensity} \quad \Lambda_T(\delta, S) := \int_0^T \lambda(S_t - (S_0 - \delta))dt$$

where

- $0 \leq \delta \leq \delta_{\max}$ with $\delta_{\max} \in (0, S_0)$ denotes the *depth of the limit order book*,
- $(S_t)_{t\geq 0}$ is a stochastic process modeling the dynamic of the *best ask price of a security stock*,
- the function λ is defined on the whole real line as a finite non increasing convex function. We will use $\lambda(x) = Ae^{-ax}$.

Estimation of $\lambda(\delta)$

Let $(\tau_n^{(\delta)})_{n\geq 1}$ be the sequence of trade arrivals on $[0, T]$ at a distance δ of the best limit assumed to be r.v. i.i.d. with exponential distribution of parameter $\lambda(\delta)$.

We estimate $\lambda(\delta)$ with the estimator $\hat{\lambda}_n(\delta)$ defined by

$$\hat{\lambda}_n(\delta) = \frac{\sum_{k=1}^{n} \mathbb{1}_{\{\tau_k^{(\delta)} < T\}}}{\sum_{k=1}^{n} \tau_k^{(\delta)} \wedge T}.$$

This estimator converges *a.s.* toward $\lambda(\delta)$ for each δ and satisfies a CLT, namely

$$\hat{\lambda}_n(\delta) \xrightarrow[n\to\infty]{a.s.} \lambda(\delta) \quad \text{and} \quad \sqrt{n}\left(\hat{\lambda}_n(\delta) - \lambda(\delta)\right) \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, \sigma_\delta).$$

Example on market data

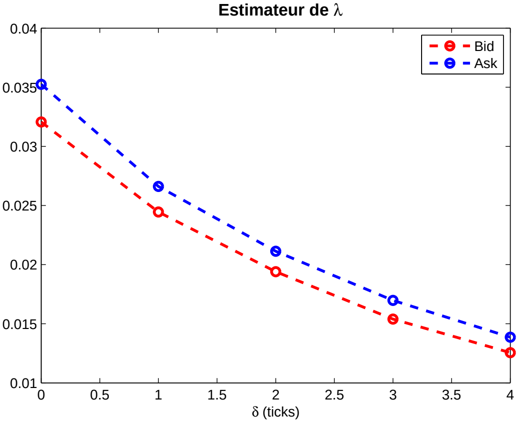

Figure: $\hat{\lambda}(\delta)$ on the bid and ask sides for ACCOR on january 2012 for 5 limits and $T = 15$ trades.

Estimation of A and a

Assume now that the execution time depends on the posting distance with the following parametrization: $\hat{\lambda}_n(\delta) = Ae^{-a\delta}$. We look for estimating the parameters A and a from the estimator $\hat{\lambda}_n(\delta)$.

We will minimize the prediction error with the following linear regression

$$\log \hat{\lambda}_n(\delta) = c + \beta\delta + \epsilon.$$

Then

$$\hat{A} = \exp(\hat{c}) \quad \text{and} \quad \hat{a} = -\hat{\beta}.$$

Application on market data

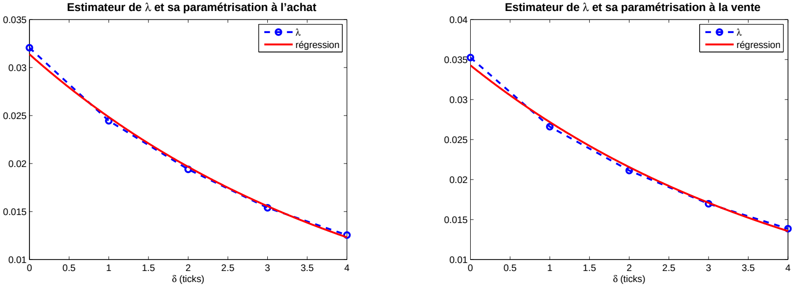

Figure: Comparison of parametrizations of estimators $\hat{\lambda}$ at the bid and the ask sides for ACCOR on january 2012 obtained by linear regression: $A_{buy} = 0.0314$, $a_{buy} = 0.2339$, $A_{sell} = 0.0343$, $a_{sell} = 0.2318$.

Outline

Design of the cost function

- Over the period $[0, T]$, we aim at executing a portfolio of size $Q_T \in \mathbb{N}$ invested in the asset S.
- At time $t = 0$, a buy order of size Q_T is posted at price $S_0 - \delta$.
- On $[t, t + \Delta t]$, the *probability for such an order to be executed* is $\lambda(S_t - (S_0 - \delta))\Delta t = A \exp\left(-a\left(S_t - (S_0 - \delta)\right)\right)\Delta t$.
- The *execution cost* for a distance δ is $\mathbb{E}\left[(S_0 - \delta)\left(Q_T \wedge N_T^{(\delta)}\right)\right]$.
- We add to this execution cost a **penalization** depending on the *remaining quantity* to execute $\left(Q_T - N_T^{(\delta)}\right)_+$, namely at the end of the period T, we want to have Q_T assets in the portfolio.

Optimization Problem

Then we introduce a **market impact penalization function** $\Phi : \mathbb{R} \mapsto \mathbb{R}_+$, nondecreasing and convex, with $\Phi(0) = 0$ to model the additional cost of the execution of the remaining quantity.

Then the *resulting cost of execution* on a period $[0, T]$ reads

$$C(\delta) := \mathbb{E}\left[(S_0 - \delta)\left(Q_T \wedge N_T^{(\delta)}\right) + \kappa S_T \Phi\left(\left(Q_T - N_T^{(\delta)}\right)_+\right)\right]$$

where $\kappa > 0$.

Our aim is then to **minimize this cost**, namely to solve the following optimization problem

$$\min_{0 \le \delta \le \delta_{\max}} C(\delta).$$

To solve this optimization problem, we will devise a **stochastic algorithm constrained** to stay in $[0, \delta_{\max}]$.

Strategy

We will devise a **stochastic algorithm constrained** to stay in $[0, \delta_{\max}]$. To this end we have to

- prove that C, C' and C'' admit representations as expectations, *i.e.* in particular, there exists a Borel functional

$$H : [0, \delta_{\max}] \times \mathbb{D}([0,T], \mathbb{R}) \longrightarrow \mathbb{R} \quad \text{such that}$$

$$\forall \delta \in [0, \delta_{\max}], \quad C'(\delta) = \mathbb{E}\left[H\left(\delta, (S_t)_{t\in[0,T]}\right)\right]$$

- find natural assumptions on Q_T and κ such that C is twice differentiable, strictly convex on $[0, \delta_{\max}]$ with $C'(0) < 0$. Consequently

$$\text{argmin}_{\delta\in[0,\delta_{\max}]} C(\delta) = \{\delta^*\}, \quad \delta^* \in (0, \delta_{\max}]$$

  and

$$\delta^* = \delta_{\max} \text{ iff } C \text{ is non increasing on } [0, \delta_{\max}].$$

Design of the algorithm

Once the two points are checked, we can devise the algorithm following the **standard stochastic approximation with projection**, namely

$$\delta_{n+1} = \mathrm{Proj}_{[0,\delta_{\max}]}\left(\delta_n - \gamma_{n+1} H\left(\delta_n, \left(\bar{S}_{t_i}^{(n+1)}\right)_{0\le i\le m}\right)\right),\ \delta_0 \in [0, \delta_{\max}],$$

where

- $\mathrm{Proj}_{[0,\delta_{\max}]}$ denotes the projection on $[0, \delta_{\max}]$,
- the positive step sequence $(\gamma_n)_{n\ge1}$ satisfies at least the minimal *decreasing step* assumption

$$\sum_{n\ge1} \gamma_n = +\infty \quad \text{and} \quad \gamma_n \to 0,$$

- $\left\{\left(\bar{S}_{t_i}^{(n)}\right)_{0\le i\le m}, n \ge 0\right\}$ is either a sequence of i.i.d. copies of the true underlying dynamics of $(S_{t_i})_{0\le i\le m}$ (see [2, 3, 4]) or at least of its Euler scheme or a sequence sharing some averaging properties (*e.g.* stationary α-mixing, see [5]).

Outline

Basic facts on Poisson distributed variables

Let $(N^\mu)_{\mu>0}$ be a family of Poisson distributed random variables with parameter $\mu > 0$.

(i) For every function $f : \mathbb{N} \to \mathbb{R}_+$ such that $\log f(n) = O(n)$,

$$\frac{d}{d\mu}\mathbb{E}\left[f(N^\mu)\right] = \mathbb{E}\left[f(N^\mu + 1) - f(N^\mu)\right] = \mathbb{E}\left[f(N^\mu)\left(\frac{N^\mu}{\mu} - 1\right)\right].$$

In particular, for any $k \in \mathbb{N}$, $\frac{d}{d\mu}\mathbb{P}\left(N^\mu \le k\right) = -\mathbb{P}\left(N^\mu = k\right)$.

For any $k \in \mathbb{N}^*$,

(ii) $\mathbb{E}\left[k \wedge N^\mu\right] = k\mathbb{P}\left(N^\mu > k\right) + \mu\mathbb{P}\left(N^\mu \le k-1\right)$ and $\frac{d}{d\mu}\mathbb{E}\left[k \wedge N^\mu\right] = \mathbb{P}\left(N^\mu \le k-1\right)$,

(iii) $\mathbb{E}\left[(k - N^\mu)_+\right] = k\mathbb{P}\left(N^\mu \le k\right) - \mu\mathbb{P}\left(N^\mu \le k-1\right)$,

(iv) $k\mathbb{P}\left(N^\mu = k\right) = \mu\mathbb{P}\left(N^\mu = k-1\right)$.

Representation as expectations of C and C'

To compute the cost function (or its gradient), it is convenient to use a pre-conditioning with respect to $\mathcal{F}_T^S := \sigma\left(S_t, 0 \le t \le T\right)$. We come down to compute the above quantity when $N^{(\delta)}$ is replaced by a standard Poisson random variable with parameter μ denoted N^μ.

$$C(\delta) = \mathbb{E}\left[\widetilde{C}\left(\delta, \Lambda_T(\delta, S), (S_t)_{0\le t\le T}\right)\right],$$

where for every $x \in \mathcal{C}([0,T], \mathbb{R}_+)$ and every $\mu \in \mathbb{R}_+$,

$$\begin{aligned}\widetilde{C}\left(\delta, \mu, x\right) &= (x_0 - \delta)\left(Q_T \mathbb{P}\left(N^\mu > Q_T\right) + \mu \mathbb{P}\left(N^\mu \le Q_T - 1\right)\right) \\ &\quad + \kappa x_T \mathbb{E}\left[\Phi\left(\left(Q_T - N^\mu\right)_+\right)\right].\end{aligned}$$

We introduce some notations for reading convenience: we set

$$\mathbb{P}^{(\delta)}\left(N^\mu > Q_T\right) = \mathbb{P}\left(N^\mu > Q_T\right)_{|\mu = \Lambda_T(\delta, S)}$$

and

$$\mathbb{E}^{(\delta)}\left[f(\mu)\right] = \mathbb{E}\left[f(\mu)\right]_{|\mu = \Lambda_T(\delta, S)}.$$

Representation as expectations of C and C'

- If $\Phi \neq \mathrm{id}$, then $C'(\delta) = \mathbb{E}\left[H(\delta, S)\right]$ with

$$
\begin{aligned}
H(\delta, S) &= -Q_T \mathbb{P}^{(\delta)}\left(N^\mu > Q_T\right) - \kappa S_T \frac{\partial}{\partial \delta} \Lambda_T(\delta, S) \varphi^{(\delta)}(\mu) \\
&\quad + \left(\frac{\partial}{\partial \delta} \Lambda_T(\delta, S)(S_0 - \delta) - \Lambda_T(\delta, S)\right) \mathbb{P}^{(\delta)}\left(N^\mu \leq Q_T - 1\right)
\end{aligned}
$$

where $\varphi^{(\delta)}(\mu) = \mathbb{E}^{(\delta)}\left[\left(\Phi\left(Q_T - N^\mu\right) - \Phi\left(Q_T - N^\mu - 1\right)\right) \mathbb{1}_{\{N^\mu \leq Q_T - 1\}}\right]$.

- If $\Phi = \mathrm{id}$, then $C'(\delta) = \mathbb{E}\left[H(\delta, S)\right]$ with

$$
\begin{aligned}
H(\delta, S) &= -Q_T \mathbb{P}^{(\delta)}\left(N^\mu > Q_T\right) \\
&\quad + \left((S_0 - \delta - \kappa S_T) \frac{\partial}{\partial \delta} \Lambda_T(\delta, S) - \Lambda_T(\delta, S)\right) \mathbb{P}^{(\delta)}\left(N^\mu \leq Q_T - 1\right).
\end{aligned}
$$

Criteria for the monotony at the origin and convexity

Assume that $(S_t)_{t\in[0,T]}$ satisfies a functional co-monotony principle. To simplify, set $\lambda(x) = Ae^{-ax}$, $A, a > 0$.

(a) **Monotony at the origin:** $C'(0) < 0$ as soon as

$$Q_T \geq 2T\lambda(-S_0) \quad \text{and} \quad \kappa \leq \frac{1 + aS_0}{a\mathbb{E}\left[S_T\right]\left(\Phi(Q_T) - \Phi(Q_T - 1)\right)}.$$

In particular, when $\Phi \equiv \mathrm{id}$, the condition reduces to

$$\kappa \leq \frac{1 + aS_0}{a\mathbb{E}\left[S_T\right]}.$$

(b) **Convexity.** If $\Phi \neq \mathrm{id}$, assume that Φ satisfies

$$\forall x \in [1, Q_T - 1], \quad \Phi(x) - \Phi(x-1) \leq \rho_Q(\Phi(x+1) - \Phi(x)), \quad \rho_Q \in (0,1).$$

$C''(\delta) \geq 0$, $\delta \in [0, \delta_{\max}]$ (so that C is convex on $[0, \delta_{\max}]$) if

$$Q_T \geq 2T\lambda(-S_0) \quad \text{and} \quad \kappa \leq \frac{2}{a\mathbb{E}\left[S_T\right]\Phi'_\ell(Q_T)}.$$

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Market data

As market data, we use the bid prices of Accor SA (ACCP.PA) of 11/11/2010 for the fair price process $(S_t)_{t\in[0,T]}$. We divide the day into periods of 15 trades which will denote steps of the stochastic procedure. Let N_{cycles} be the number of these periods. For every $n \in \{1,\dots,N_{\text{cycles}}\}$, we have a sequence of bid prices $(\bar{S}^{(n)}_{t_i})_{0\le i\le 14}$ and we approximate the jump intensity of the Poisson process $\Lambda_{T^n}(\delta,S)$, where $T^n = \sum_{i=0}^{14} t_i$, by

$$\forall n \in \{1,\dots,N_{\text{cycles}}\}, \quad \Lambda_{T^n}(\delta,S) = A\sum_{i=1}^{14} e^{-a(\bar{S}^{(n)}_{t_i} - \bar{S}_{t_0} + \delta)}(t_i - t_{i-1}).$$

The empirical mean of the intensity function

$$\bar{\Lambda}(\delta,S) = \frac{1}{N_{\text{cycles}}}\sum_{n=1}^{N_{\text{cycles}}} \Lambda_{T^n}(\delta,S)$$

is plotted on the following figure.

Intensity function

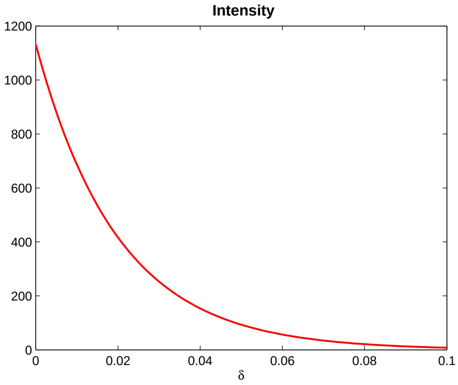

Figure: Fit of the exponential model on real data (Accor SA (ACCP.PA) 11/11/2010): $A = 1/50$, $a = 50$ and $N_{\text{cycles}} = 220$.

The penalization function is of the following form

$$\Phi(x) = (1 + \eta(x))x \quad \text{with} \quad \eta(x) = A' e^{a' x}.$$

Cost function and its derivative

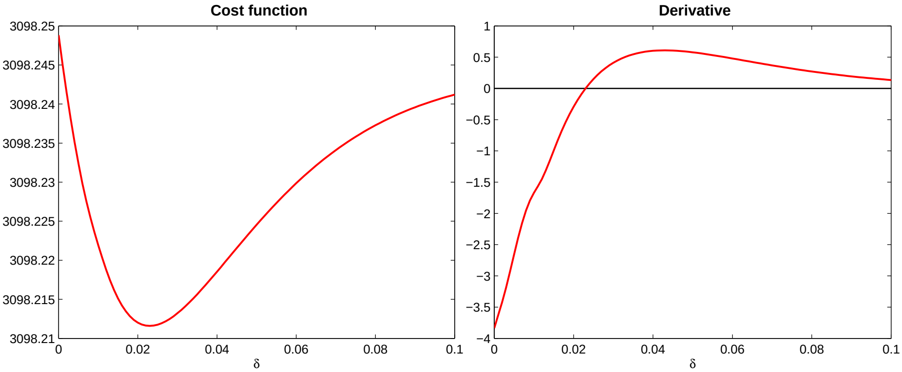

Figure: $\eta \not\equiv 0$: $A = 1/50$, $a = 50$, $Q = 100$, $\kappa = 1$, $A' = 0.001$, $a' = 0.0005$ and $N_{\mathrm{cycles}} = 220$.

δ and posting price obtained by SA

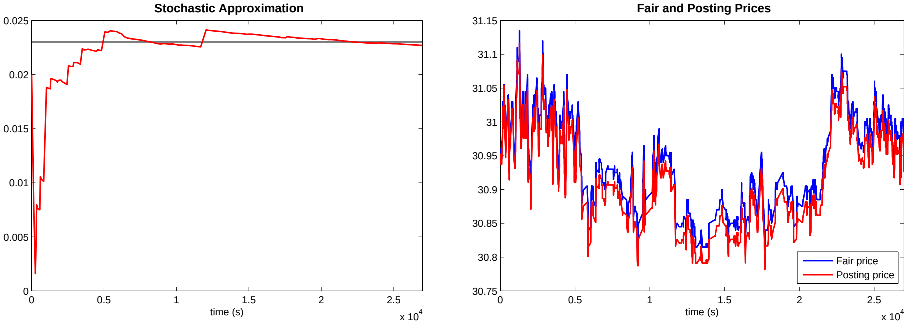

Figure: $\eta \not\equiv 0$: $A = 1/50$, $a = 50$, $Q = 100$, $\kappa = 1$, $A' = 0.001$, $a' = 0.0005$ and $N_{\text{cycles}} = 220$. Crude algorithm with $\gamma_n = \frac{1}{550n}$.

Stochastic algorithm with RP averaging

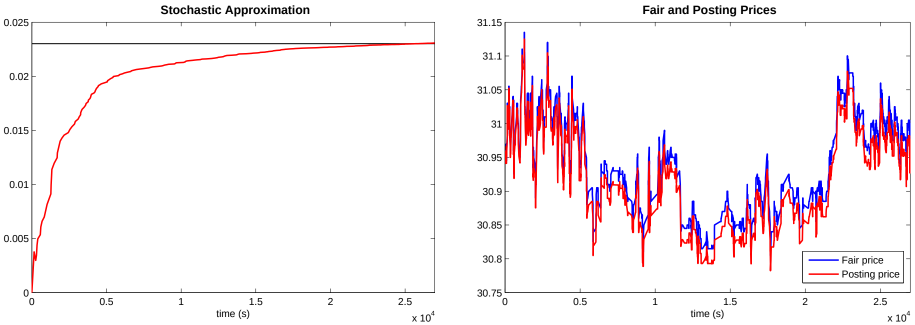

Figure: $\eta \not\equiv 0$: $A = 1/50$, $a = 50$, $Q = 100$, $\kappa = 1$, $A' = 0.001$, $a' = 0.0005$ and $N_{\text{cycles}} = 220$. Averaging algorithm (Ruppert and Poliak) with $\gamma_n = \frac{1}{550 n^{0.95}}$.

Reminder. Averaging algorithm (Ruppert and Poliak) $\bar{\delta}_n = \frac{1}{n+1} \sum_{k=0}^{n} \delta_k$, $n \geq 1$.
The main asset of this averaging procedure is that, under appropriate assumptions, its weak rate of convergence is ruled by a *CLT* having a minimal asymptotic variance (among all recursive procedures).

Implementation

- Estimate on market data for one asset the intensity function λ depending on the first 5 limits.
- Estimate the coefficients A and a of the parametrization $\lambda(x) = Ae^{-ax}$.
- Fix T (in number of trades) and Q_T (approximately 1 ATS). Compute the cost function and its derivative using the previous parametrization (with the parameters given by the linear regression). You will have to tune κ to be sure that you have a convex cost.
- Implement the stochastic algorithm and check it converges toward the minimum (you could use averaging principle).

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